

Benjamin Knepper

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EDUCATION

University of California, Berkeley, College of Letters & Science

Berkeley, CA

Majors: Physics (with honors), Philosophy; *Minor:* Mathematics

08/2023 - present

Physics GPA: 4.0/4.0

B.A. in progress for 12/2025

Cornell University, College of Arts & Sciences

Ithaca, NY

Majors: Physics, Philosophy

08/2020 - 05/2022

Cumulative GPA: 3.77/4.0

RESEARCH EXPERIENCE

UC Berkeley Physics Honors Bachelor's Thesis

Berkeley, CA

Theoretical Student Researcher

03/2025 - current

- Calculating the “standard quantum limit” (SQL) of measuring the length of a Jackiw-Teitelboim gravity wormhole in the Hartle-Hawking state, using open quantum system linear response theory
- Identified the Sachdev-Ye-Kitaev (SYK) dual operator of wormhole length via chord diagrams and Krylov complexity in the triple-scaled limit, and now computing bulk-boundary correlation functions
- Developing both digital (qubit) and analog (cold atom cavity) experimental implementations

Advanced Quantum Testbed (AQT)

Berkeley, CA

Theoretical and Experimental Research Collaborator

08/2024 - current

- Deriving a protocol to measure an effective “Page curve” in chaotic many-body quantum systems, relating quantum Fisher information with out-of-time-order correlators (OTOCs)
- Wrote a successful research proposal to simulate this measurement on AQT superconducting qutrits, with quantum information scrambling generated by pseudorandom unitaries
- Coding OTOC simulations in Google Cirq software for an experimental test underway in Fall 2025

Lawrence Berkeley National Laboratory (LBNL)

Berkeley, CA

QuIPS (Quantum Invisible Particle Sensor) Experiment Researcher

11/2023 - current

- Collaborating on Geant4 Monte Carlo simulations of unique first-forbidden β^- decays and analyses of signal-to-noise projections for the upcoming QuIPS beyond-standard-model search
- Designed an optimal configuration of particle physics calorimeters surrounding an optomechanically levitated nanosphere, achieving a QuIPS detector sensitivity to a sterile neutrino coupling of 10^{-3}

Network for Neutrinos, Nuclear Astrophysics, and Symmetries (N3AS)

Berkeley, CA

Theoretical Student Researcher

08/2023 - 12/2023

- Initiated a phenomenological model to detect ultra-light dark matter from primordial black holes using asteroseismology and observations of oscillations in relativistic stars

University of Chicago Enrico Fermi Institute

Chicago, IL

UChicago Temporary Research Professional

05/2023 - 08/2023

- Measured blackbody spectrum of the GigaBREAD experiment with a Vector Network Analyzer to determine system noise temperature, confirming sensitivity to $O(10)$ GHz resonant modes
- Performed Fourier signal analysis of the thermal noise which identified a contaminant background
- Wrote Python data acquisition scripts for the current best dark photon limit of 10^{-12} in $[10.7, 12.5]$ GHz

Fermi National Accelerator Laboratory

Batavia, IL

D.O.E SULI (Science Undergraduate Laboratory Internships) Intern

01/2023 - 05/2023

- Performed optical ray-tracing simulations in FRED software to characterize focusing errors of signal

photons onto a Superconducting Nanowire Single Photon Detector (SNSPD) in InfraBREAD

- Designed a novel photon detector configuration involving a Winston Cone and reflector optics that improves quantum sensing efficiency by 55%, even with misalignments from cryogenic cooling

Millennium Institute of Astrophysics

Santiago, Chile

Research Assistant

06/2022 - 09/2022

- Modeled the transit photometry and radial velocity data of a NASA TESS exoplanet candidate in Python, leading to the verification of the novel warm jupiter planet TOI-6628b
- Coded with Bayesian parameter optimization algorithms such as Markov Chain Monte Carlo (MCMC)

PUBLICATIONS

- Peer Reviewed:*
1. G. Hoshino et al. (GigaBREAD Collaboration), “First Axion-Like Particle Results from a Broadband Search for Wave-Like Dark Matter in the 44 to 52 μeV Range with a Coaxial Dish Antenna,” [*Phys.Rev.Lett.* **134** 171002](#) (2025).
 2. M. Tala Pinto et al., “Three Warm Jupiters and one sub-Saturn orbiting TOI-6628, TOI-3837, TOI-5027 and TOI-2328,” [*Astronomy & Astrophysics* **694** A268](#) (2025).
 3. S. Knirck et al. (BREAD Collaboration), “First Results from a Broadband Search for Dark Photon Dark Matter in the 44 to 52 μeV Range with a Coaxial Dish Antenna”, [*Phys.Rev.Lett.* **132** 131004](#) (2024), *Editor’s Suggestion*.
- Manuscripts:*
4. B. Knepper, A. Sonnenschein, S. Knirck, “Focusing Optics for Axion Detection: Simulating Sensing Enhancements of Photons in InfraBREAD,” [*OSTI* **2377355**](#) (2024).

HONORS & AWARDS

- Recipient of the national 2024 American Institute of Physics, Society of Physics Students [“Outstanding Undergraduate Research Award”](#), \$500

SELECT CONFERENCES & WORKSHOPS

- 2025 Quantum Sensing and Precision Science Summer School, Johns Hopkins, MD — *participant*
- 2025 APS Global Summit March-April Meeting, Anaheim, CA — *talk*
- 2025 Observables in Quantum Gravity: From Theory to Experiment, Aspen Center for Physics, CO — *poster presentation*
- 2024 Vienna Quantum Foundations Conference, Institute for Quantum Optics and Quantum Information (IQOQI), Vienna, Austria — *poster presentation*
- 2023 SQMS Annual Ecosystem Day Collaboration Meeting, Fermilab, IL — *poster presentation*
- 2023 US QIS Summer School, Superconducting Materials and Systems Center (SQMS), IL — *participant*

TECHNICAL SKILLS

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| <ul style="list-style-type: none">• General research: programming, numerical simulations, data analysis, formal presentations• Fields: quantum information theory, quantum metrology, quantum optics, nuclear physics, quantum nonequilibrium and chaos theory• Software: Python, Cirq, Qiskit, C++, Geant4 Monte Carlo, Mathematica, GitHub, FRED Photon Engineering, Linux, LaTeX | <ul style="list-style-type: none">• Theoretical: squeezed states, sub-SQL measurements, input-output theory, SYK, JT gravity, holographic bulk reconstruction, thermal correlators, random matrix theory• Experimental: superconducting qubits, beyond standard model searches, optomechanical quantum sensors, Radio-Frequency (RF) measurements, cryogenics, CCD and CMOS calorimeters |
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